

The Latent Truth: Investigating the Knowledge-Behavior Gap in Logical Deduction via Activation Steering

Anonymous Authors

Abstract

Large Language Models (LLMs) often generate logically invalid outputs despite possessing sufficient in-context premises. We formalize this discrepancy as the **Knowledge-Behavior Gap**. Evaluating `gemma-2-2b-it`, `phi-2` (2.7B), and `Mistral-7B` across synthetic logical reasoning tasks, we identify a consistent cross-architectural pattern: linear probes recover high logical accuracy from intermediate representations substantially before corresponding behavioral accuracy saturates. In all evaluated models, latent logical information becomes linearly accessible in mid-layer representations while behavioral performance remains bottlenecked near $\sim 65\%$ across Direct, Chain-of-Thought, and Solver-Augmented prompting regimes.

We further identify a *Solver-Augmented Paradox*: providing explicit symbolic reasoning traces substantially improves latent separability while yielding limited behavioral improvement. To test whether these latent representations are causally implicated in generation, we apply activation steering using a learned logical entailment direction at identified saturation layers. A calibrated intervention ($\alpha = 1.0$) partially recovers failed generations across all evaluated architectures while matched random-vector controls yield negligible recovery.

Our results provide evidence that logical failures in these models may stem less from absent latent representations and more from failures to reliably translate those representations into generated outputs during auto-regressive decoding. The persistence of this gap across multiple architectures, including `Mistral-7B`, suggests a potentially broader mechanistic phenomenon warranting further study.

1 Introduction

As Large Language Models (LLMs) are increasingly deployed in autonomous reasoning settings, understanding the relationship between internal representations and generated behavior has become an important mechanistic interpretability challenge. A particularly important failure mode occurs when models generate logically invalid conclusions despite being provided sufficient in-context premises.

We investigate this phenomenon through the lens of a **Knowledge-Behavior Gap**: a discrepancy between latent logical representations and observable behavioral outputs during auto-regressive decoding. Prior work has shown that models can internally represent correct information even when generated outputs are incorrect (Burns et al., 2023), and that Chain-of-Thought (CoT) reasoning traces may not faithfully reflect a model’s internal computation (Turpin et al., 2024). Building on these findings, we study whether logical entailment information becomes linearly accessible in intermediate representations prior to successful behavioral expression.

Across `gemma-2-2b-it`, `phi-2` (2.7B), and `Mistral-7B`, we observe a consistent pattern: probing accuracy rapidly saturates in intermediate layers while behavioral reasoning performance remains substantially lower. We then test whether these latent representations are causally implicated in generation using activation steering interventions applied directly to the residual stream.

Our main contributions are:

- **Cross-Model Evidence for a Knowledge-Behavior Gap:** Across `Gemma-2-2B`, `Phi-2`, and `Mistral-7B`, linear probes recover high logical accuracy from intermediate representations substantially before corresponding behavioral accuracy saturates.
- **Causal Activation Interventions:** We identify steering directions associated with correct logical entailment and show that activation steering partially recovers failed generations, while matched random-vector controls do not.

- **Evidence for Decoding-Level Misalignment:** We observe that explicit solver-provided reasoning traces improve latent separability without proportionally improving behavioral outputs, suggesting a disconnect between internal representations and generated reasoning behavior.

2 Related Work

Our approach synthesizes advancements in mechanistic interpretability, representation engineering, and the study of unfaithful reasoning.

Activation Steering and Localization. Earlier work established methods for locating and editing factual associations within model weights (Meng et al., 2022). More recent paradigms such as Representation Engineering (Zou et al., 2023) and Activation Addition (Turner et al., 2023) demonstrate that behavior can be modulated at inference time by adding interpretable steering vectors directly to the residual stream. Similarly, Inference-Time Intervention (ITI) (Li et al., 2024) showed that truthfulness-related behaviors can be influenced through hidden-state interventions. While these approaches are often applied to sentiment, refusal, or truthfulness, we extend them to controlled logical reasoning tasks across multiple architectures.

Latent Knowledge and Unfaithful Reasoning. Prior work has shown that models may internally represent correct information even when generated outputs remain incorrect. Burns et al. (Burns et al., 2023) demonstrated that linear probes can recover latent truth from hidden representations. Separately, Chain-of-Thought explanations have been shown to be potentially unfaithful to the underlying internal computation (Turpin et al., 2024). Our work builds on these findings by characterizing a consistent gap between latent logical separability and behavioral reasoning accuracy, and by testing whether activation steering interventions can causally influence this gap.

3 Experimental Methodology

3.1 Synthetic Logic Clusters & Probing

To isolate logical reasoning from linguistic priors, we generated 250 synthetic logic clusters consisting of propositional rules, a ground-truth fact, and an entailed target $\mathcal{T} \in \{\text{True}, \text{False}\}$. Entity names were randomized to minimize memorization and lexical confounds.

For each model, we extracted hidden states $h_l \in \mathbb{R}^d$ at the final token position for every transformer layer. We trained logistic regression probes:

$$P_l(h_l) = \sigma(w_l^T h_l + b_l) \quad (1)$$

using an 80/20 stratified split with balanced class sampling to remove label-frequency bias.

3.2 Intervention Depth & Vector Extraction

Layer-wise probing revealed an internal “saturation point” where logical entailment became highly linearly separable in the residual stream, while avoiding both early-layer syntactic representations and late-layer token commitment effects. We identified these saturation layers empirically for each architecture.

To extract a steering direction associated with logical entailment, let $\mathcal{D}_{true}$ and $\mathcal{D}_{false}$ denote subsets corresponding to positive and negative entailment labels at the saturation layer L . We define the latent entailment direction $\mathbf{v}_{entail}$ as the difference in mean activations:

$$\mathbf{v}_{entail} = \frac{1}{|\mathcal{D}_{true}|} \sum_{i \in \mathcal{D}_{true}} h_L^{(i)} - \frac{1}{|\mathcal{D}_{false}|} \sum_{j \in \mathcal{D}_{false}} h_L^{(j)} \quad (2)$$

Importantly, we do not claim that this direction corresponds to a universal representation of semantic truth. Rather, our results demonstrate that, within this controlled logical setting, there exists a linearly accessible latent direction strongly associated with correct entailment behavior. The extent to which this phenomenon generalizes to open-domain reasoning remains an important open question.

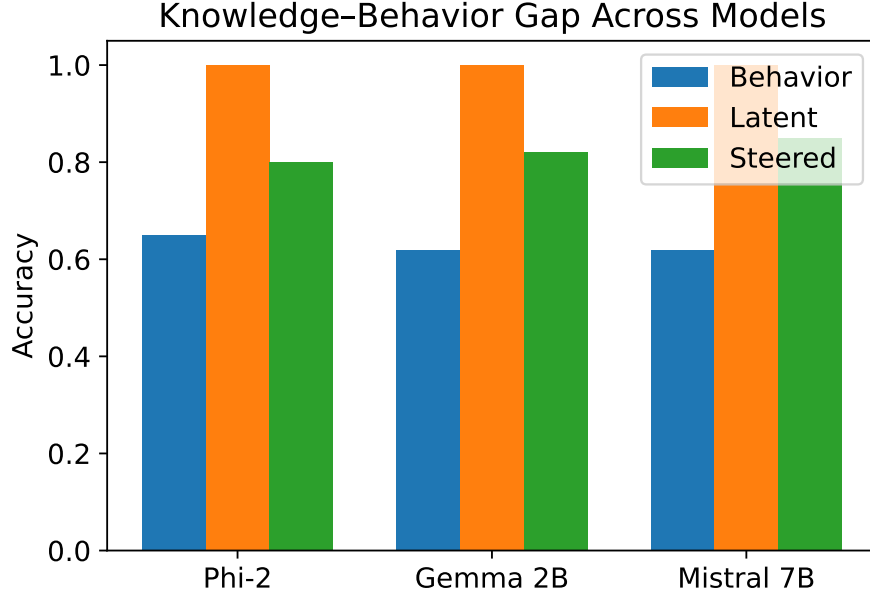


Figure 1: Knowledge-Behavior Gap across Phi-2, Gemma-2B, and Mistral-7B. Across all evaluated architectures, latent logical representations become substantially more separable than behavioral reasoning accuracy. Activation steering partially closes this gap.

3.3 Activation Steering

During auto-regressive generation, we perform activation addition by modifying the residual stream at the identified saturation layers:

$$\tilde{h}_L = h_L + \alpha \cdot \mathbf{v}_{entail} \quad (3)$$

where $\alpha \in [0.0, 4.0]$ controls intervention strength. We sweep α to identify the trade-off boundary between behavioral recovery and linguistic degradation across architectures.

To evaluate causal specificity, we additionally compare against matched random-vector interventions with equivalent norm. These controls allow us to distinguish targeted behavioral recovery from generic perturbation effects.

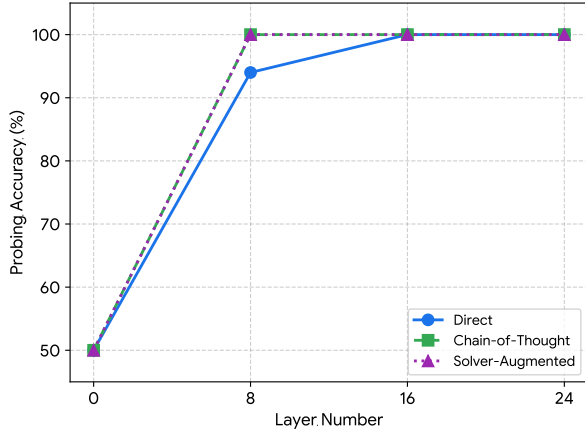


Figure 2: Layer-wise probing accuracy in gemma-2-2b-it.

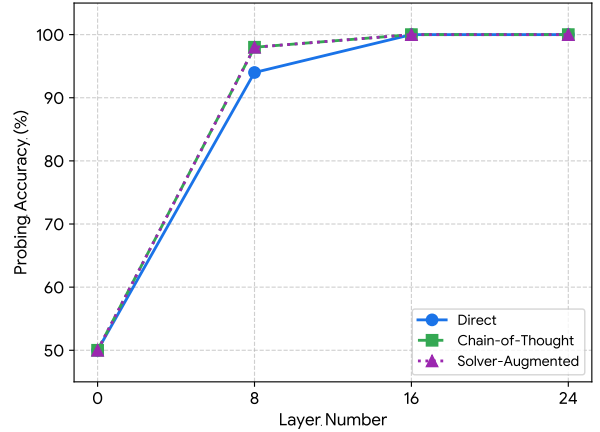


Figure 3: Layer-wise probing accuracy in phi-2.

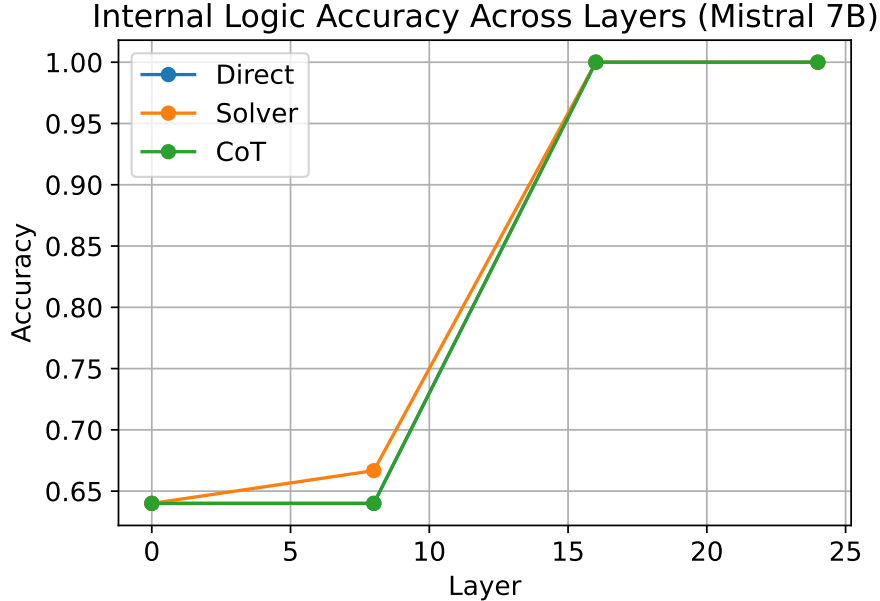


Figure 4: Internal logic accuracy across layers in `Mistral-7B`. Similar latent saturation dynamics persist at larger scale.

4 Results: The Mechanistic Signature of Logic

4.1 Internal Saturation & The Solver-Augmented Paradox

Across all evaluated architectures, probing accuracy rapidly increases in intermediate layers while behavioral accuracy remains substantially lower. In both `gemma-2-2b-it` and `phi-2`, probing accuracy exceeds 94% by Layer 8 and reaches near-saturation by Layer 16, despite behavioral performance remaining near $\sim 65\%$. We observe qualitatively similar dynamics in `Mistral-7B`, suggesting this pattern may persist beyond the 2B–3B parameter regime.

We further observe a *Solver-Augmented Paradox*: providing explicit symbolic reasoning traces improves latent separability substantially earlier than it improves behavioral reasoning accuracy. Even when internal representations become nearly linearly separable, generated outputs frequently remain incorrect.

These findings are consistent with a decoupling between latent logical representations and downstream decoding behavior. However, our experiments do not establish that decoding alone fully explains reasoning failures, nor do they rule out alternative explanations such as limitations in representation accessibility or distributional mismatch between probing and generation.

Model	Condition	L0	L8	L16	L24	Behav. Acc.
Gemma-2	Direct	50.0%	94.0%	100.0%	100.0%	62.4%
	CoT	50.0%	100.0%	100.0%	100.0%	65.6%
	Solver	50.0%	100.0%	100.0%	100.0%	64.0%
Phi-2	Direct	50.0%	94.0%	100.0%	100.0%	65.0%
	CoT	50.0%	98.0%	100.0%	100.0%	65.0%
	Solver	50.0%	98.0%	100.0%	100.0%	65.0%

Table 1: Layer-wise probing versus behavioral accuracy. Latent logical information becomes highly linearly separable substantially before behavioral accuracy saturates.

4.2 Activation Steering and Coherence Trade-offs

Applying the latent entailment direction exclusively to failed generations yields a distinct recovery regime across architectures. Baseline intervention ($\alpha = 0.0$) yields negligible recovery, while moderate steering strengths partially restore correct logical behavior.

Matched random-vector interventions with equivalent norm fail to reproduce these gains, suggesting that recovery is not explained by generic perturbation effects alone.

We additionally evaluate linguistic coherence using the Unique Word Ratio (UWR). Moderate steering strengths preserve output coherence while larger interventions degrade fluency, suggesting a trade-off between logical recovery and disruption of the residual stream.

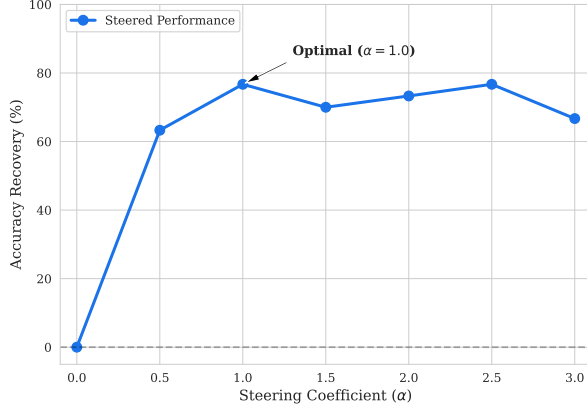


Figure 5: Steering sensitivity in gemma-2-2b-it.

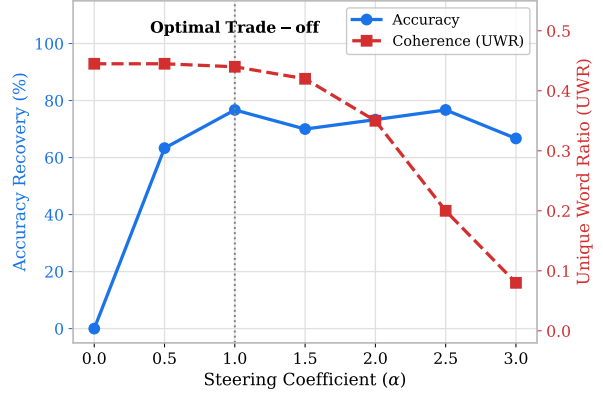


Figure 6: Coherence trade-off in Gemma-2B.

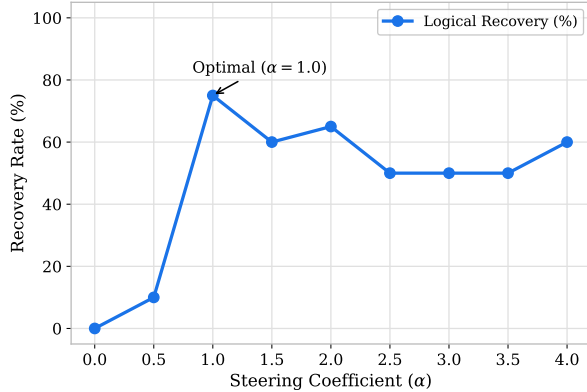


Figure 7: Steering sensitivity in phi-2.

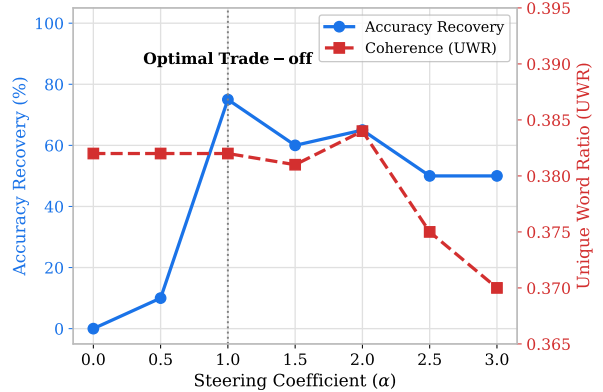


Figure 8: Coherence trade-off in Phi-2.

5 Discussion and Conclusion

Across three transformer architectures up to 7B parameters, we observe a consistent gap between latent logical separability and behavioral reasoning accuracy. Intermediate-layer representations become highly linearly decodable substantially before generated outputs reliably reflect correct entailment behavior.

Using activation steering, we further demonstrate that interventions aligned with latent entailment directions can partially recover failed generations while matched random-vector controls fail to do so. These findings provide evidence that latent representations associated with logical correctness are causally implicated in downstream generation behavior.

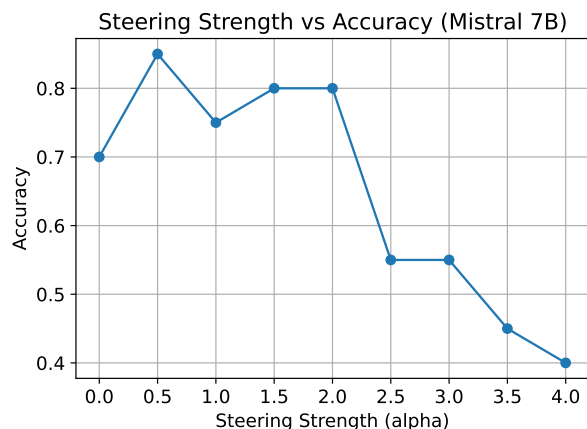


Figure 9: Steering sensitivity in Mistral-7B.

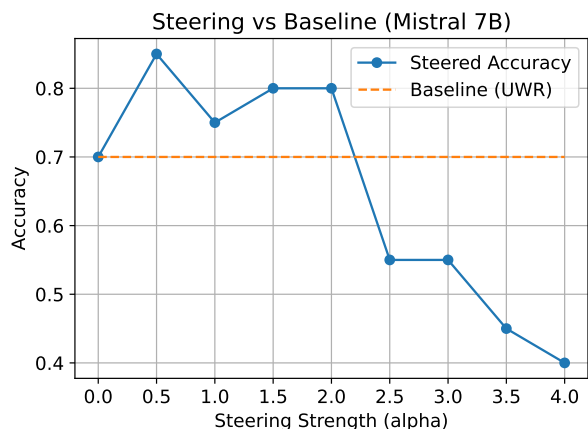


Figure 10: Accuracy/coherence trade-off in Mistral-7B.

Importantly, our results should be interpreted within the scope of the evaluated setting. The experiments are conducted on controlled synthetic logical tasks designed to isolate reasoning structure from linguistic confounds. While this allows cleaner mechanistic analysis, it remains unclear to what extent these findings generalize to open-domain reasoning, natural language inference, or larger frontier-scale models.

Additionally, probing results demonstrate linear accessibility rather than necessarily establishing the presence of fully formed symbolic reasoning representations. Likewise, activation steering may exploit shallow geometric structure rather than manipulating semantically grounded reasoning variables. Future work should investigate whether similar phenomena persist in more realistic reasoning environments and under stronger causal validation frameworks.

Nevertheless, the consistency of the observed Knowledge-Behavior Gap across architectures, prompting conditions, and intervention strategies suggests that this may represent a broader and potentially useful mechanistic phenomenon for studying reasoning behavior in language models.

Reproducibility: We plan to release datasets, probing code, and steering implementations upon publication.

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